

2 Eye colour in humans is an example of polygenic inheritance.

Two of the genes giving rise to this characteristic are found on chromosome 15.

(a) There is more than one version of each of the genes involved.

Which term is used for different versions of a gene?

(1)

- ☐ **A** allele
- ☐ **B** locus
- ☐ **C** phenotype
- ☐ **D** polygenic

(b) Inheritance of different versions of these two genes shows genetic linkage.

Explain how the phenotypes of an individual can show genetic linkage in the inheritance of two genes.

(2)

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- (c) One gene can give rise to more than one type of protein through post-transcriptional changes to the mRNA.

One of these changes is the removal of introns.

Explain how different proteins can be synthesised from this mRNA that has had introns removed.

(3)

(Total for Question 2 = 6 marks)

